

Safety Compliant Control for Robotic Manipulator With Task and Input Constraints

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Abstract—Increasingly, robots are working in close proximity with humans, and in dynamically changing environments. We present a new control architecture that guarantees safety under such conditions, while simultaneously ensuring that task goals are satisfied. Our approach combines Control Barrier Functions (to provide safety guarantees) with Rapidly Exponentially Stabilizing Control Lyapunov functions (to satisfy task constraints and ensure exponential convergence). We formulate the problem completely in the operational space, using super ellipsoids to define safe sets which are dynamically adapted to restrict the robot's operational space in response to moving obstacles (including humans). Control barrier functions, which can be implemented in real time using quadratic programming, ensure the forward invariance of these safe sets, while minimally perturbing prescribed control trajectories. We demonstrate our approach on a seven degree-of-freedom Kuka LBR iiwa robotic manipulator, both in a physics engine-based simulation and with real hardware experiments.

Index Terms—Compliance and impedance control, constrained motion planning, collision avoidance, robot safety.

I. INTRODUCTION

WITH advances in computation, automation, and robotics, interest in the autonomous operation of robotic manipulators has expanded beyond the specially contrived environments that were once typical of factory automation, moving into applications where the environment is dynamic and uncertain, and in which humans move freely (e.g., logistics, construction [1]–[4]). These new applications require that robots be able to move among dynamic obstacles, including humans, while avoiding collision, and providing safety guarantees.

The problem of collision avoidance in robotics has been a focus of robotics research for many years [5]–[7], either by constructing collision-free motion plans [8]–[10], or by incorporating obstacle avoidance directly into the control architecture to ensure safety [11]–[15]. In this paper, we take the latter approach, and present a method that modifies the robot's control

input in real time to achieve task goals while providing provable safety guarantees.

Our approach is to formulate safe sets in the operational space, and to then design control laws that ensure forward invariance of these sets. As obstacles (including other robots or humans) move through the robot's workspace, these safe sets are dynamically modified by restricting the robot's operational space. We use Control Barrier Functions (CBFs) [16] to ensure forward invariance of the safe sets [17], and Rapidly Exponentially Stabilizing Control Lyapunov Functions (RESCLFs) [18] to ensure that task goals are achieved.

CBFs have been used for differential drive robots [19], quadrotors [20], [21], and bipedal locomotion [22]. They have recently been applied to the problem of collision avoidance using an operational space formulation in [12], [23]. RESCLFs can be used to achieve exponential convergence while computing a min-norm control law, and they have been applied to problems as complex as controlling bipeds with hybrid dynamics [18]. A unified controller with RESCLF and CBF was used in [24], both being implemented using Quadratic Programming (QP), which can be computed in real-time.

We model safe sets in the operational space as super ellipsoids; these can be considered as virtual cages for the manipulator. These higher-order ellipsoids define the control barrier functions in the operational space. We encode obstacle avoidance using additional CBFs to ensure that the manipulator avoids obstacles present within its operational space. To the best of our knowledge, this is the first paper to address the problem of restricting the operational space of the robotic manipulator with the resulting controller satisfying both safety and tracking constraints.

The main contributions of our paper can be summarized as follow:

- Defining safe sets in the operational space using super ellipsoids.
- Formulating these operational space constraints as CBF constraints to yield real-time solutions while satisfying obstacle avoidance and tracking constraints.
- Exploiting the forward invariance properties of safe sets even when the end-effector is initially outside the safe set.
- Unifying the CBF constraints with the RESCLF framework to yield safety and tracking in joint formulation.

We show the efficacy of our proposed approach on KUKA LBR iiwa 7 DoF robotic manipulator, both in drake physics engine [25] and actual robotic platform.

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The remainder of the paper is organized as follows. Section II describes the operational space for the manipulator, control Lyapunov functions, and control barrier functions in operational space. Section III defines super ellipsoids and illustrates how they can be used to define safe sets in the operational space, and to ensure their forward invariance. This section also provides a unified formulation to compute rapidly exponentially stabilizing control Lyapunov functions and safe set constraints using control barrier function, along with any additional task constraints. We discuss simulation and experimental results in section IV, and we conclude the paper in section V.

II. BACKGROUND AND MATHEMATICAL FORMULATION

We will first briefly describe system modeling, followed by RESCLF and CBFs in operational space.

A. System Model

The system dynamics of the manipulator is given as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \Gamma \quad (1)$$

where $M \in \mathbb{R}^{N \times N}$ is the mass/inertia matrix, $C \in \mathbb{R}^{N \times N}$ is the Coriolis term, $G \in \mathbb{R}^N$ is the gravitational term, and $\Gamma \in \mathbb{R}^N$ are the applied torques to the system. $q \in \mathbb{R}^N$, $\dot{q} \in \mathbb{R}^N$ and $\ddot{q} \in \mathbb{R}^N$ represents the joint position, velocity and acceleration of the manipulator. For sake of simplicity, we will refer mass, coriolis and gravitational term as M , C , and G only. The operational space of the manipulator can be written as

$$\begin{aligned} \dot{x} &= J(q)\dot{q} \\ \ddot{x} &= \dot{J}(q, \dot{q})\dot{q} + J(q)\ddot{q} \end{aligned} \quad (2)$$

where J and $\dot{J}$ are the jacobian and jacobian derivative. x , and $\dot{x}$ represents the operational space of the end-effector of the manipulator. Let the domain of the operational space be represented by $\mathcal{T}$. For example, if x represents the position of the end-effector, then $\mathcal{T} = \mathbb{R}^3$ for position and $\dot{x}$ represents the linear velocity of the end-effector. We can then link operational space to the system model of a manipulator by substituting $\ddot{q}$ in (2). This yields [12]

$$\begin{aligned} \ddot{x} &= \dot{J}\dot{q} + JM^{-1}[\Gamma - C(q, \dot{q})\dot{q} - G(q)] \\ &= \underbrace{\dot{J}\dot{q} + JM^{-1}[-C(q, \dot{q})\dot{q} - G(q)]}_{f(q)} + \underbrace{JM^{-1}\Gamma}_{g(q)} \end{aligned} \quad (3)$$

We can also represent system dynamics in terms of operational space by first representing $\ddot{x}$ in terms of jacobian as

$$\ddot{q} = J^\dagger (\ddot{x} - \dot{J}\dot{q}) \quad (4)$$

where $J^\dagger$ is the pseudo inverse of the jacobian. We can then substitute it into system equation to yield

$$MJ^\dagger (\ddot{x} - \dot{J}\dot{q}) + C(q, \dot{q})\dot{q} + G(q) = \Gamma \quad (5)$$

The benefit of the above formulation will be clear when they will be used in conjugation with RESCLF and CBFs in operational space.

B. Rapidly Exponentially Stabilizing Control Lyapunov Function in Operational Space

In order to apply RESCLF, we first consider the (4) and feedback linearize the system by applying the following input:

$$\Gamma = MJ^\dagger (\ddot{x}^{des} - \dot{J}\dot{q}) + C(q, \dot{q})\dot{q} + G(q) \quad (6)$$

where $\ddot{x}^{des}$ is the desired acceleration. In simplest form, we can define $\ddot{x}^{des}$ on error function as

$$\ddot{x}^{des} = -K_p e - K_v \dot{e} \quad (7)$$

where $e = x - x^{des}$ is the error function on position, and $\dot{e} = \dot{x} - \dot{x}^{des}$ is the error function on velocity. K_p and K_v are positive definite matrices. Applying this input will yield the following

$$\ddot{x} = \ddot{x}^{des}. \quad (8)$$

In order to incorporate input constraint and ensure exponential convergence where applicable, we introduce an auxiliary input, μ , into the actual input, Γ , as follow:

$$\Gamma = MJ^\dagger (\ddot{x}^{des} + \mu - \dot{J}\dot{q}) + C(q, \dot{q})\dot{q} + G(q). \quad (9)$$

This will yield $\ddot{x} = \ddot{x}^{des} + \mu$. Let $\ddot{e} = \ddot{x} - \ddot{x}^{des}$, then we can write

$$\ddot{e} = \mu. \quad (10)$$

We can write error dynamics in normal form by first defining $\tilde{\eta} = [e; \dot{e}]$. We can then write error dynamics as [26]

$$\dot{\tilde{\eta}} = F\tilde{\eta} + G\mu \quad (11)$$

where F and G are given as

$$F = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ I \end{bmatrix} \quad (12)$$

and 0 and I are zero and identity matrices. We can then define a Lyapunov candidate $V = \tilde{\eta}^T P \tilde{\eta}$, where P is the positive definite matrix and is obtained by solving algebraic Riccati equation as follow:

$$F^T P + PF - PGG^T P + Q = 0 \quad (13)$$

where Q is positive definite symmetric matrix. We can then take the time derivative of V to yield

$$\dot{V} = \underbrace{\tilde{\eta}^T (F^T P + PF) \tilde{\eta}}_{L_F V} + \underbrace{2\tilde{\eta}^T PG \mu}_{L_G V}. \quad (14)$$

where $L_F V$ and $L_G V$ represents the Lie derivative of $V(\tilde{\eta})$. RESCLF can then be obtained by satisfying the following inequality

$$L_F V + L_G V \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\tilde{\eta}) \quad (15)$$

where $\lambda_{\min}(Q)$ and $\lambda_{\max}(P)$ are the minimum and maximum eigenvalues of Q and P matrices respectively. We can satisfy the RESCLF constraint for both position and orientation as well

as input constraint by formulating it as a quadratic optimization problem given as [18]

$$\mu^* = \arg \min_{\mu} \mu^T \mu$$

subject to :

$$\begin{aligned} L_F V(\tilde{\eta}) + L_G V(\tilde{\eta}) \mu &\leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\tilde{\eta}) \\ L_F V^{Or}(\tilde{\eta}) + L_G V^{Or}(\tilde{\eta}) \mu &\leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V^{Or}(\tilde{\eta}) \\ M J^\dagger (\ddot{x}_{des} + \mu - \dot{J} \dot{q}) + C(q, \dot{q}) \dot{q} + G(q) &\leq \Gamma_{max} \\ -M J^\dagger (\ddot{x}_{des} + \mu - \dot{J} \dot{q}) - C(q, \dot{q}) \dot{q} - G(q) &\leq \Gamma_{max} \end{aligned} \quad (16)$$

where V^{Or} is the Lyapunov function associated with orientation. The benefit of representing orientation as a separate constraint will be evident when we want to maintain safety and orientation while avoiding obstacles. Next, we will briefly describe CBFs in operational space.

C. Control Barrier Functions in Operational Space

Control barrier functions are used to ensure safety by first designing a safe set of states and then utilizing control barrier functions to guarantee forward invariance of the given set, i.e., if the system starts in the safe set, the system will remain in the safe set. For safety in operational space, the safe set must be also defined in operational space. Control barrier functions ensure safety by modifying the nominal controller in a minimally invasive manner.

We can define safe set for manipulator in terms of its operational space. If x represents the operational space, and $\mathcal{C}_0$ is the safe set such that barrier function, $h(x) \geq 0$, ensures safety. Then we can define it as follows

$$\mathcal{C}_0 = \{x \in \mathcal{T} \mid h(x) \geq 0\} \quad (17)$$

For most robotics system including manipulators, input, Γ , appears by taking the second derivative of the (17) as

$$\ddot{h}(x) = L_f^2 h(x) + L_g L_f h(x) \Gamma \quad (18)$$

where $L_f^2 h(x)$ and $L_g L_f h(x)$ represent the Lie derivatives for $h(x)$. This also implies that barrier function, $h(x)$, has a relative degree 2. We can then utilize a variant of control barrier function called Exponential Control Barrier Function (ECBF) [27]. ECBF states that we can ensure safety by satisfying the following inequality

$$L_f^2 h(x) + L_g L_f h(x) \Gamma + K \eta \geq 0 \quad (19)$$

where $K \in \mathbb{R}^{1 \times 2}$, and $\eta = [h(x) \ \dot{h}(x)]$. The coefficient of vector, K , can be selected by the considering it as a problem of pole placement for the closed loop matrix $[F - GK]$ such that the eigenvalues, λ , of the closed loop matrix are placed at $\lambda = -[p_0, p_1]^T$ and p_0, p_1 are positive and real. If we define $v_0 = h(x)$ and $v_i = \dot{v}_{i-1} + p_i v_0$, then each pole must also satisfy $p_{i-1} \geq -\frac{\dot{v}_i(x_0)}{v_i(x_0)} \forall i \in \{0, 1\}$, where x_0 is the initial

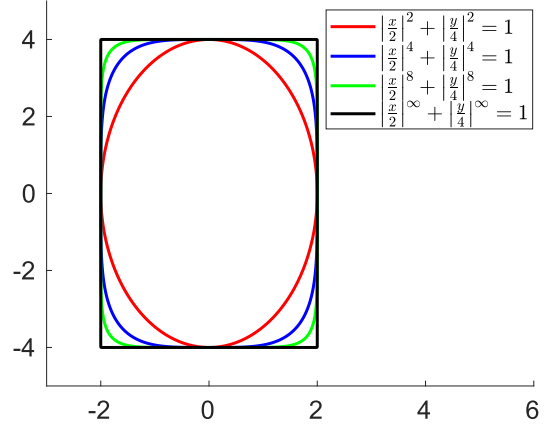


Fig. 1. Higher Order Ellipsoid and its approximation to perfect rectangle.

condition of the operational space [27]. If Γ^{nom} is the nominal control input, then we can utilize QP formulation to yield safe controller as

$$\Gamma^* = \arg \min_{\Gamma} \frac{1}{2} \|\Gamma - \Gamma^{nom}\|^2$$

subject to :

$$L_f^2 h(x) + L_g L_f h(x) \Gamma + K \eta \geq 0 \quad (20)$$

We can then apply the resulting controller, Γ^* , to ensure forward invariance of the safe set. Next, we will define safe set for constraining the operational space of the robotic manipulator.

III. OPERATIONAL CONSTRAINT

In order to ensure safety, we propose to limit the operational space of the robotic manipulators. Inspired by the classical design of limiting the manipulators in physical cages, we propose to limit the operational space of robotic manipulators in virtual boxes or virtual cuboids. This can be done by considering approximating the dimensions of virtual cuboids by the interior of a higher-order ellipsoid, also called super-ellipsoids [28]. A higher-order ellipsoid perfectly approximates a cuboid as the power approaches infinity. However, infinity is unrealizable in practice, and the function is also not smooth. A visual 2D representation of higher-order ellipsoid is given in Fig. 1.

If $y_c = [x_{1c} \ x_{2c} \ x_{3c}]$ with x_{1c}, x_{2c} , and x_{3c} represents a center point in the world, $y = [x_1 \ x_2 \ x_3]$ represents a generic point in the world, and $A \in \mathbb{R}^{3 \times 3}$ is a positive definite symmetric matrix with values on diagonal representing inverse of length, width and height. Then the standard equation of ellipsoid can be written in matrix form as

$$[y - y_c]^T A [y - y_c] = 1. \quad (21)$$

Equation of interior of the ellipsoid in higher order is

$$\left| \frac{x_1 - x_{1c}}{length} \right|^{2n} + \left| \frac{x_2 - x_{2c}}{width} \right|^{2n} + \left| \frac{x_3 - x_{3c}}{height} \right|^{2n} \leq 1 \quad (22)$$

where n is positive even integer. However, we can also represent it in matrix form as

$$[|y - y_c|^n]^T A [|y - y_c|^n] \leq 1 \quad (23)$$

where A represents the virtual cuboid. Similarly, if x represents the operational space of the manipulator, and x_c represents the base of manipulator, then the safe set can be represented as

$$h(x) = 1 - [|x - x_c|^n]^T A [|x - x_c|^n] \quad (24)$$

where x_c represents the base of the manipulator in the world. If operational space is unconstrained, then we can also set large values for length, width and height representing the virtual cuboids. The first derivative of $h(x)$ yields

$$\begin{aligned} \dot{h}(x) = & -2 [|x - x_c|^n]^T A [\text{sign}(x - x_c) \circ \\ & \times n(x - x_c)^{n-1} \circ (\dot{x} - \dot{x}_c)] \\ & - [|x - x_c|^n]^T \dot{A} [|x - x_c|^n] \end{aligned} \quad (25)$$

where $\circ$ is the point-wise multiplication. Let $\tilde{X} = [\text{sign}(x - x_c) \circ n(x - x_c)^{n-1} \circ (\dot{x} - \dot{x}_c)]$, then the second derivative of $h(x)$ yields

$$\begin{aligned} \ddot{h}(x) = & -2\tilde{X}^T A \tilde{X} - 2 [|x - x_c|^{n/2}]^T A [\text{sign}(x - x_c) \circ \\ & \times \{n(n-1)(x - x_c)^{n-2} \circ (\dot{x} - \dot{x}_c) \circ (\dot{x} - \dot{x}_c) \\ & + n(x - x_c)^{n-1} \circ (\ddot{x} - \ddot{x}_c)\}] - 2 [|x - x_c|^n]^T \ddot{A} \tilde{X} \\ & - [|x - x_c|^n]^T \ddot{A} [|x - x_c|^n]^T. \end{aligned} \quad (26)$$

We can then plug $\ddot{x}$ from (3) in (26) to write $h(x)$ as

$$\ddot{h}(x) = L_f^2 h(x) + L_g L_f h(x) \Gamma. \quad (27)$$

Control barrier functions can then be enforced by solving the QP formulation as mentioned in (20). The approach of limiting the operational space does not limit the ability of manipulator to avoid obstacle avoidance within operational constraint. In fact, we can also utilize control barrier function for obstacle avoidance, as given in [12], and compute the solution using QP formulation as follow

$$\Gamma^* = \arg \min_{\Gamma} \frac{1}{2} \|\Gamma - \Gamma^{nom}\|^2$$

subject to :

$$\begin{aligned} L_f^2 h_c(x) + L_g L_f h_c(x) \Gamma + K_c \eta_c & \geq 0 \\ L_f^2 h_o(x) + L_g L_f h_o(x) \Gamma + K_o \eta_o & \geq 0 \end{aligned} \quad (28)$$

where h_c and h_o are the operational and obstacle constraints for the manipulator. Similarly, $\eta_c = [h_c \ \dot{h}_c]$ and $\eta_o = [h_o \ \dot{h}_o]$ and K_c and K_o is the vector associated with each quantity. We can

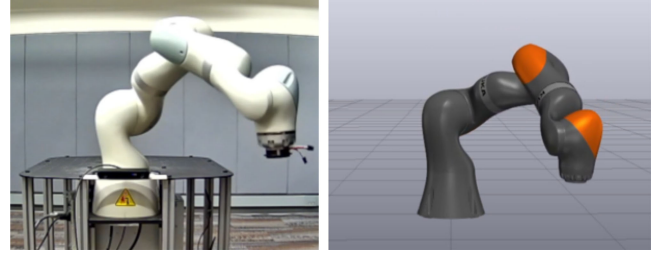


Fig. 2. Snapshot of the experiments in simulation and actual robotic platform.

also combine it with RESCLF to yield a unified controller as

$$\mu^* = \arg \min_{\mu, \delta, \Gamma} \mu^T \mu + p \delta^2$$

subject to :

$$\begin{aligned} L_F V(\tilde{\eta}) + L_G V(\tilde{\eta}) \mu & \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\tilde{\eta}) + \delta \\ L_F V^{Or}(\tilde{\eta}) + L_G V^{Or}(\tilde{\eta}) \mu & \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V^{Or}(\tilde{\eta}) \\ M J^\dagger (\ddot{x}_{des} + \mu - \dot{J} \dot{q}) + C(q, \dot{q}) \dot{q} + G(q) & \leq \Gamma_{max} \\ -M J^\dagger (\ddot{x}_{des} + \mu - \dot{J} \dot{q}) - C(q, \dot{q}) \dot{q} - G(q) & \leq \Gamma_{max} \\ \Gamma = M J^\dagger (\ddot{x}_{des} + \mu - \dot{J} \dot{q}) + C(q, \dot{q}) \dot{q} + G(q) \\ L_f^2 h_c(x) + L_g L_f h_c(x) \Gamma + K_c \eta_c & \geq 0 \\ L_f^2 h_o(x) + L_g L_f h_o(x) \Gamma + K_o \eta_o & \geq 0 \end{aligned} \quad (29)$$

where δ is the relaxation variable to ensure that solution always exists and $p > 0$ is a weight variable. The formulation only relaxes the position variable so that manipulator still maintains the orientation while avoiding obstacles or operating within operational constraints.

IV. RESULTS

We apply the framework to a 7 DoF ‘‘KUKA LBR iiwa 14 med’’ robotic manipulator firstly in drake physics engine [25], and then on the actual KUKA iiwa arm for experimental verification. KUKA has not released actual parameters for KUKA iiwa robotic manipulator, and therefore, our experimental verification is performed with approximated values. We enforced the torque limit to a fraction of the actual datasheet to show the efficacy of the framework. The maximum values in Nm in the datasheet are $[320 \ 320 \ 176 \ 176 \ 110 \ 40 \ 40]^T$ for each joint respectively, but we enforced the torque limits to $[80 \ 80 \ 60 \ 60 \ 55 \ 15 \ 15]^T$. The objective of the manipulator was as follows: a) Track a reference trajectory while maintaining the orientation of the end-effector, b) do not violate the operational space constraint, and c) maintain safety by avoiding obstacles within the operational space. We utilized quaternions to define error for orientation control [29]. QP optimization was done using [30] and the average computation time for optimization was around 276 microseconds,

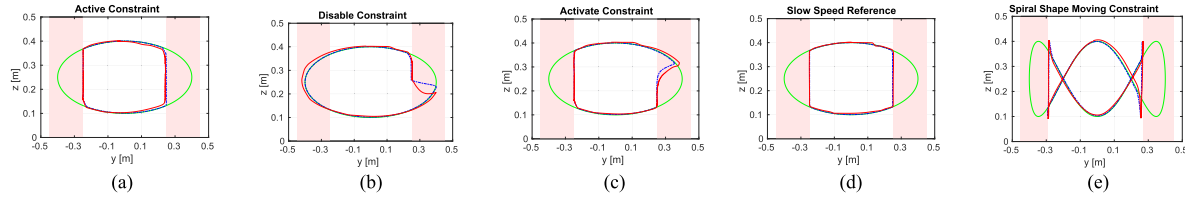


Fig. 3. End-effector trajectory while satisfying input and safety constraints. The green line is the target trajectory, the blue line is the end-effector trajectory in simulation, and the red line is the end-effector trajectory on the actual robotic platform. The pink region represents the reduced operational space barrier.

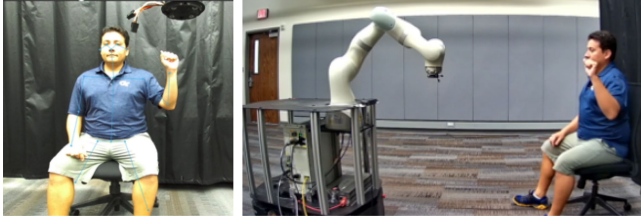


Fig. 4. Snapshot of the obstacle avoidance experiment.

and the average time for the whole computational framework was around 920 micro-second. We chose poles for the barrier to be $[360, 60]$ for obstacle avoidance and $[320, 80]$ for operational space constraint. The magnitude of the poles determines how strongly the safety of the barrier should be enforced, and they can be selected by the hit-n-trial method for practical consideration [24]. The results show that the manipulator can successfully satisfy all the constraints in simulation; however, it experienced some degradation in the actual experiments on the robotic platform. The snapshot of the experiments are shown in Fig. (2). The video illustrating the experiments is submitted along with the paper.

In order to check the efficacy of the proposed framework in terms of tracking and safety, we initially formulated the target to move in the ellipsoid in the $y - z$ direction with 0.8 m value as the major axis and 0.3 m value as the minor axis of the ellipsoid. The operational space constraint was initially set to values $[1\ 1\ 1]$ implying the length, width, and height of the ellipsoid were set to 1 m each, and the power of higher-order ellipsoid was set to $n = 8$. We initially allowed the manipulator to follow the reference trajectory, and since the reference was within the operational constraint of the barrier, we achieved excellent tracking. We then constrained the barrier to $[1\ 0.25\ 1]$, i.e., the width of the ellipsoid is reduced to 0.25 m , which implied that the reference trajectory will now leave the operational space of the manipulator, but the end-effector of the manipulator should only operate in the permissible operational space. Therefore, the end-effector of the manipulator did not follow it outside but remained within the operational space constraint such that it minimizes the error. The result is shown in Fig. (3)(a). We then expanded the barrier to $[1\ 1\ 1]$ abruptly when the reference target was outside of the safe set and the end-effector was inside the operational constraint. The expansion in the safe set causes the end-effector to converge to the target position as the

safety constraints are now removed, and the reference trajectory is now included in the operational space of the manipulator. This result can be seen in Fig. (3)(b). We next reduced the operational constraint from $[1\ 1\ 1]$ to $[1\ 0.25\ 1]$ suddenly, while the end-effector was outside the reduce safe set following the reference trajectory. The sudden reduction in the size of the safe set causes the end-effector to abruptly go to the safe set, and we also observe a large magnitude in the torques. This is because the end-effector was significantly outside the safe set. It would not have been the case if we had gradually decreased the safe set. However, we performed experiments to show the efficacy of the proposed technique, and the results are shown in Fig. (3)(c). We then performed the experiment with the slow-moving target, shown in Fig. (3)(d) and the fast-moving target while the barriers were activated. Finally, we moved the reference trajectory to make a ∞ symbol and changed the barrier gradually, which is shown in Fig. (3)(e). We observed that safety and input constraints were satisfied in all the scenarios while performing trajectory tracking. We see some degradation in performance on the actual robot, represented by the red-line in Fig. (3), especially in terms of orientation control, which is primarily due to approximated parameters. The experimental results are shown in the video submission of this paper.

Finally, we show the efficacy of the proposed technique for obstacle avoidance by performing the following experiments: a) target reference trajectory moves in a line, b) the reference trajectory moves outside the operational space, and c) an obstacle is introduced in the reference trajectory. The objective of the manipulator is to track the reference trajectory while it is within the operational constraint of the manipulator and avoid obstacles irrespective of the reference trajectory passing through the obstacle. The manipulator should still maintain orientation while satisfying the constraint. We performed body tracking for the experiment, and the obstacle is considered the person's left wrist. We assumed a virtual ellipsoid of radius 0.15 m on the manipulator's end-effector. We then pushed our wrist towards the end-effector of the manipulator to replicate the scenario of an approaching obstacle. It deviates from the reference trajectory as soon as the obstacle is observed within the safe set and restores the trajectory when the obstacle moves away. The snippet of the experiment is given in Fig. (4) and the experiment can be seen in the accompanying videos. It can also be accessed at.¹

¹[Online]. Available: <https://youtu.be/PGt7EaxXMLA> and <https://youtu.be/PGt7EaxXMLA>

V. CONCLUSION

We presented a framework to limit the operational space of the manipulator besides the obstacle avoidance problem using the control barrier function. The framework uses QP formulation to ensure that we can compute the solution in real-time while satisfying multiple constraints like obstacle avoidance, limiting operational space, and input saturation. It yields excellent tracking and generates a min-norm solution. Finally, we show the efficacy of the proposed framework on the KUKA LBR iiwa manipulator in simulation and the actual robotic platform.

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